

- Q.23 Explain working of synchromesh box in brief.  
 Q.24 State function of steering mechanism.  
 Q.25 Write a short note on single plate clutch.  
 Q.26 What are the merits and demerits of automatic transmission.  
 Q.27 Describe construction and working of torque converter.  
 Q.28 Explain Ackermann's steering mechanism  
 Q.29 Difference between dry and wet clutch  
 Q.30 Write a short note on differential and its working.  
 Q.31 Explain :  
     Cornering force  
     over steering and under steering  
 Q.32 Explain Hotchkiss drive.  
 Q.33 What are the functions of propeller shaft.  
 Q.34 Discuss common faults of transmission.  
 Q.35 Discuss Elliot and reverse Elliot types stub axle.

#### SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)  
 Q.36 What are different parts of chassis? Also describe function of each part.  
 Q.37 Describe construction and working of sliding gear box.  
 Q.38 Explain construction and working of hydraulic power steering.

(340) (4) 180343/170343/120343  
 /030343

No. of Printed Pages : 4 180343/170343/120343  
 Roll No. .... /030343

#### 4th Sem / Auto Subject:- Chassis, Body and Transmission - 1

Time : 3Hrs. M.M. : 100

#### SECTION-A

**Note:** Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Which of the following is a classification of automobiles based on Load?  
 a) Heavy transport vehicle (HTV)  
 b) Sedan Hatchback car  
 c) Four Wheeler vehicle  
 d) Front-wheel drive  
 Q.2 The function of the alternator  
 a) Recharging the Battery  
 b) Voltage Regulator  
 c) Auto-ignition  
 d) None of the above  
 Q.3 The purpose of the reciprocating ball type steering gear.  
 a) To reduce the operating cost  
 b) To reduce the number of parts  
 c) To reduce the operating friction  
 d) To reduce the toe-out during the turns

(1) 180343/170343/120343  
 /030343

- Q.4 Which types of joints are used when the shaft are inclined?  
 a) Universal joint      b) Hinge joint  
 c) Ball and Socket joint d) Pivot joint
- Q.5 Which of the following is the need of the gearbox?  
 a) To vary the speed of the vehicle  
 b) To vary the torque of the vehicle  
 c) To vary the power of the vehicle  
 d) To vary the acceleration of the vehicle
- Q.6 Why are the helical gears used commonly in transmission over spur gears?  
 a) Low cost and high strength  
 b) Low noise level and high strength  
 c) Low noise level and economy  
 d) Low noise level and low cost
- Q.7 The following type of arrangement is used in synchromesh type gear box  
 a) Single plate clutch  
 b) Fluid clutch  
 c) Dog clutch  
 d) Semi-centrifugal clutch
- Q.8 A transfer case is used  
 a) All Vehicles  
 b) Front wheel drive vehicle  
 c) Rear wheel drive vehicle  
 d) Four Wheel drive vehicle

(2) 180343/170343/120343  
 /030343

- Q.9 The torque which a clutch can transmit, depends upon the  
 a) Coefficient of friction  
 b) Spring force  
 c) Contact Surfaces  
 d) all of the above
- Q.10 The component of the torque converter that allows multiplication of torque is the  
 a) Turbine                      b) Impeller  
 c) Pump                         d) Stator

### SECTION-B

**Note:** Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 The lubricant used in gear box is called \_\_\_\_\_
- Q.12 Give name of any two types of gear box.
- Q.13 Define fluid drive
- Q.14 Define Toe in.
- Q.15 Write one application of live axle.
- Q.16 Define chassis.
- Q.17 What is the function of clutch.
- Q.18 The function of slip joint is \_\_\_\_\_
- Q.19 Wet clutch is also known as \_\_\_\_\_
- Q.20 LMV stand for.

### SECTION-C

**Note:** Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Give detail of front axle.
- Q.22 Write down necessity of automobile.

(3) 180343/170343/120343  
 /030343